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Calibration of $PM_{2.5}$ mass concentrations used in the Pittsburgh Aerosol Research and Inhalation Epidemiology Study

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HIGHLIGHTS

- >75,000 daily $PM_{2.5}$ mass concentrations from 97 samplers at 47 monitoring sites.
- Measurements from 7 programs: AQS, IMPROVE, CASTNet, UORVP, PAQS, SCAMP, NETL/OST.
- Calibrations were developed using linked measurement error models.
- TEOM bias relative to FRM follows a well-defined sigmoidal temperature dependency.
- FRM was most precise while TEOM imprecision decreases with increasing temperature.

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ABSTRACT

Fifteen different types of $PM_{2.5}$ mass concentration samplers were used by seven different monitoring networks at 47 locations in the Pittsburgh, Pennsylvania, region from 1999 to 2008. The samplers included Federal Reference Method (FRM) samplers, speciation samplers, tapered element oscillating microbalance (TEOM) samplers, and others. The different measurement principles used in these designs tended to lead to systematic differences (biases) when measuring the same quantity, and to differences in the typical size of random errors (imprecision) introduced by each type of sampler. Bias can take different forms either as a constant bias or as a non-constant (scale) bias, which depends on the size of the quantity being measured. The objective of the work presented here was to simultaneously calibrate the measurements made by these different samplers to remove relative biases (both constant and non-constant) so that all of the available $PM_{2.5}$ data could be used interchangeably to develop exposure estimates for a retrospective epidemiology study. In order to accomplish this, we used linked temperature-stratified structural equation models, nonlinear regression models, and nonlinear mixed effects models. Applying these methods we constructed a comprehensive measurement error model that included both systematic error and random error components, and derived calibration equations that can be applied to place all of the $PM_{2.5}$ mass concentration measurements on the same scale. The FRM sampler was used as the reference scale although the parameter estimates are invariant to this choice. Results showed that: (1) 50 °C TEOM samplers tended to show a large downward bias relative to the FRM sampler at low temperatures, and the magnitude of this bias decreased according to a nonlinear (sigmoidal) pattern with increasing temperature, (2) speciation samplers and other integrated samplers generally showed smaller biases relative to the FRM sampler that were not temperature-dependent, and (3) FRM samplers tended to be more precise than non-FRM samplers. These results are consistent with our previous work focusing on just a single monitoring site. Results are also presented here for several types of samplers that were not part of our previous study, including 30 °C TEOM, FDMS TEOM, and beta attenuation monitors.

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1. Introduction

The goal of the Pittsburgh Aerosol Research and Inhalation Epidemiology Study (PARIES) is to determine the extent to which

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human health outcomes are associated with ambient concentrations of fine particulate matter ($PM_{2.5}$), its chemical components, and gaseous co-pollutants in the Pennsylvania counties of Allegheny (which includes the city of Pittsburgh), Washington, and Westmoreland. As part of this study, $PM_{2.5}$ mass concentration data were obtained from 97 samplers located at 47 monitoring sites in the 35 counties surrounding and including the three-county study area, in an effort to fully utilize all available information on the temporal and spatial variability of $PM_{2.5}$ to derive the best possible exposure estimates and reduce exposure misclassification error (Talbot et al., 2009; Haley et al., 2009; Bravo et al., 2012). These samplers collectively provided more than 75,000 24-h average $PM_{2.5}$ mass concentration measurements over the 10-year study period from 1999 to 2008.

The 97 samplers consisted of 15 different types based on designs including the Federal Reference Method (FRM), speciation, dichotomous, beta attenuation, and tapered element oscillating microbalance (TEOM). These types of samplers are prone to differ in systematic ways, and none provides a perfectly accurate measurement of the true ambient concentration of $PM_{2.5}$. For example, it is well known that TEOM $PM_{2.5}$ measurements tend to differ from FRM measurements, and the nature of the difference varies with temperature (Allen et al., 1997; Ayers et al., 1999; Chung et al., 2001; Rizzo et al., 2003; Charron et al., 2004; Lee et al., 2005). The FRM sampler also is prone to biases relative to true ambient $PM_{2.5}$ concentrations because of the loss of semivolatile material and the effects of particle-bound water (Hering and Cass, 1999; Ashbaugh and Eldred, 2004; Jansen et al., 2002; Rees et al., 2004). Nevertheless, we chose to define the FRM as the reference sampler for purposes of presenting our calibration results because of its superior precision and widespread use relative to the other sampler types available for our study, and because it is the measure upon which compliance with the National Ambient Air Quality Standard for $PM_{2.5}$ is based in the United States. (As such, the results of our epidemiology study can best be interpreted within a regulatory context if they are expressed on an FRM-equivalent basis).

In order to remove bias between any two devices, it is necessary to be able to simultaneously estimate the calibration equation relating the two measurements and determine the imprecision (typical size of the random error) for each device. This requires having suitable collocated measurements. In this study, a comprehensive measurement error model was constructed making it possible to determine the calibration equation linking any two monitor types. The basic methodology was described for a single monitoring site in our previous article (Bilonick et al., 2015). This methodology is extended here to simultaneously calibrate measurements made from all 97 samplers at the 47 monitoring sites used in PARIES.

2. $PM_{2.5}$ mass concentration data and measurement methods

The $PM_{2.5}$ mass concentration data used in PARIES were collected by seven ambient air quality monitoring programs that operated in the 35-county study region during some or all of the 1999–2008 period. These networks included the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS) (US EPA, 2010), the Interagency Monitoring of Protected Visual Environments (IMPROVE) network (Interagency Monitoring of Protected Visual Environments, 2010), the Clean Air Status and Trends Network (CASTNET) (Clean Air Status and Trends Network, 2010), the Upper Ohio River Valley Project (UORVP) (Upper Ohio River Valley Project, 2015), the Pittsburgh Air Quality Study (PAQS) (Pittsburgh Air Quality Study, 2015), the Steubenville Comprehensive Air Monitoring Program (SCAMP) (Steubenville Ccomprehensive Air Monitoring Program, 2015), and the National Energy Technology Laboratory Office of Science and Technology (NETL/OST) monitoring site (National Energy Technology Lab, 2015). Data from AQS, IMPROVE, and CASTNET were obtained online from the web sites for each of these monitoring networks; data from UORVP, PAQS, SCAMP, and NETL/OST were obtained directly from the researchers affiliated with those programs.

As stated above, these seven monitoring programs collectively operated 97 $PM_{2.5}$ samplers, including 15 different types of samplers, at 47 monitoring sites in the study region during 1999–2008. Table 1 provides details on the 15 sampler types, and Table 2 provides information on the 47 monitoring sites used in PARIES, including the location of each site, the types of samplers used at each site, and the number of observations (24-h average $PM_{2.5}$ mass concentrations) from each sampler type at each site. Site locations are also depicted on a map in Fig. 1. Throughout the rest of this manuscript, the Sampler Type numbers in Table 1 and the Site ID numbers in Table 2 are used as the primary keys for identifying sampler designs and monitoring sites, respectively.

The 15 $PM_{2.5}$ sampler types used in PARIES include six different models of FRM samplers (Sampler Types 1, 2, 3, 4, 5, and 8 in Table 1). These samplers, by definition, were manufactured to meet the design and performance requirements set forth in the FRM. (Reference Method for the Determination of Fine Particulate Matter as $PM_{2.5}$ in the Atmosphere, 1997) As indicated in Section 3.1, they are all treated as a single sampler type (FRM) for the remainder of this paper, and the other nine sampler types are each calibrated using the FRM as a reference. These other sampler types include four types of $PM_{2.5}$ speciation samplers (Sampler Types 6, 9, 10, and 12 in Table 1), a dichotomous sampler (Sampler Type 7 in Table 1), a beta attenuation monitor (BAM) (Sampler Type 17 in Table 1), and three types of TEOM samplers (Sampler Types 14, 16, and 19 in Table 1).

As with the FRM samplers, the speciation and dichotomous samplers collect integrated $PM_{2.5}$ samples on Teflon filters from a stream of sampled air at ambient temperature and pressure, and the mass of $PM_{2.5}$ collected is then determined gravimetrically in the laboratory under controlled conditions of temperature and humidity. Unlike the FRM, however, these samplers are designed to simultaneously collect multiple sample splits for purposes of determining mass concentrations of different PM size fractions or subsequently determining $PM_{2.5}$ chemical speciation, and as a result have certain design characteristics (e.g., sampler inlet, particle size separator, sampling flow rate, flow control scheme, etc.) that deviate from the FRM requirements and can cause differences in performance.

The BAM and the TEOMs are designed to monitor $PM_{2.5}$ mass concentration in the field on an hourly or continuous basis. The BAM does this by measuring beta ray attenuation through a $PM_{2.5}$ sample collected on a glass fiber filter tape, and the TEOM does it by measuring the change in frequency of oscillation as $PM_{2.5}$ is collected on a Teflon-coated glass fiber filter atop an oscillating hollow tube. The three types of TEOM samplers used in PARIES differ in their operating temperature (30 °C or 50 °C) and their inclusion or exclusion of a Filter Dynamics Measurement System (FDMS), which provides diffusion drying of the sampled air and adds a second, chilled purge filter (maintained at 4 °C) to better account for measurement of semi-volatile material. These differences in design and operation can affect the extent of biases resulting from moisture (i.e., condensation and particle-bound water) and from the loss of semi-volatile particulate matter (e.g., ammonium nitrate, semi-volatile organic compounds), which is expected to be most severe at elevated operating temperatures (i.e., above 50 °C) and least severe with the FDMS.

The native data obtained from the 97 $PM_{2.5}$ monitors used in PARIES included 634,130 observations representing sampling intervals ranging from 10 min (in the case of certain TEOM measurements) to 24 h. Prior to aggregation, all data were manually reviewed for quality, and the native data validation flags applied by the data collection agencies were translated to a common set of PARIES validation flags to provide for consistent treatment in the

Table 1
Description of $PM_{2.5}$ mass concentration samplers from the seven monitoring networks used in PARIES.

Type	Description	Model	Filter	Inlet	Resolution
1	PM2.5 Federal Reference Method Sampler	R&P Partisol-Plus Model 2025 Sequential Air Sampler	Teflon	WINS Impactor	24 h
2	PM2.5 Federal Reference Method Sampler	Andersen RAAS 2.5-300 $PM_{2.5}$ Sequential Sampler	Teflon	WINS Impactor	24 h
3	PM2.5 Federal Reference/ Equivalent Method Sampler	R&P Model 2000 $PM_{2.5}$ Air Sampler with BGI VSCC	Teflon	Very Sharp Cut Cyclone	24 h
4	PM2.5 Federal Reference/ Equivalent Method Sampler	R&P Model 2025 $PM_{2.5}$ Sequential Sampler with BGI VSCC	Teflon	Very Sharp Cut Cyclone	24 h
5	PM2.5 Federal Reference/ Equivalent Method	Unspecified	Teflon	Unspecified	24 h
6	PM2.5 Speciation Sampler	Desert Research Institute Sequential Filter Sampler	Teflon	Cyclone	6-24 h
7	Dichotomous Sampler	Andersen Dichotomous Sampler	Teflon	Impactor–virtual/ concentrator	24 h
8	PM2.5 Federal Reference Method Sampler	R&P Partisol FRM	Teflon	Impactor–virtual/ concentrator	24 h
9	PM2.5 Speciation Sampler	IMPROVE Module A	Teflon	Cyclone	24 h
10	PM2.5 Speciation Sampler	Met One SASS Speciation Sampler	Teflon	Very Sharp Cut Cyclone	24 h
12	PM2.5 Speciation Sampler	CASTNET Sampler	Teflon	Cyclone	24 h
14	TEOM – 50 °C Operation	Rupprecht & Patashnick Tapered Element Oscillating Microbalance - 50 °C	Teflon-Coated Glass Fiber	Sharp Cut Cyclone	Continuous
16	TEOM – 30 °C Operation	Rupprecht & Patashnick Tapered Element Oscillating Microbalance - 30 °C	Teflon-Coated Glass Fiber	Sharp Cut Cyclone	Continuous
17	Beta Attenuation Monitor	Met One Beta Attenuation Monitor	Glass Fiber Tape	Sharp Cut Cyclone	1 h
19	TEOM – FDMS	Rupprecht & Patashnick Tapered Element Oscillating Microbalance with Filter Dynamics Measurement System	Teflon-Coated Glass Fiber	Sharp Cut Cyclone	Continuous

integrated PARIES air monitoring database. Any data that were qualified as "invalid" in the native datasets (including null records) also were qualified as "invalid" in the PARIES database; this affected 14,447 of the 634,130 observations. In addition, 8183 observations were determined to be invalid during the PARIES quality review process because of irregular sampling times or sampled volumes, inconsistent or suspicious information in the native data file (without a supporting explanation), or missing information to convert from standard conditions to local conditions. All of these invalid data were excluded from the analysis, as were 116 outliers identified in the native datasets and 126 outliers identified during the PARIES quality review. (Because there were so few outliers - less than 0.04% of the total observations – their inclusion or exclusion was not expected to materially affect the results of our analysis).

The remaining 611,258 observations included 588,746 valid measurements, as well as 22,512 measurements that were flagged in the native datasets or during the PARIES quality review but were still included in our analysis because they lacked a sufficient, documented reason for removal. These valid and flagged data were converted from standard temperature and pressure to local conditions, if necessary, and any data collected at intervals shorter than 24 h were aggregated to compute 24-h averages for comparison with the other data. These 24-h averages were considered to be valid only if they included at least 19 h of valid measurement time. (In most cases, 24-h averages were computed from midnight to midnight, but in some cases they were computed from noon to noon or from 9:00 am to 9:00 am to comport with collocated integrated measurements). This resulted in the 75,927 valid 24-h average observations shown in Table 2.

3. Nonlinear measurement error model

3.1. Building blocks

In our previous study (Bilonick et al., 2015), the classic measurement error model was thoroughly described and was applied to calibrate collocated measurements made by multiple $PM_{2.5}$ samplers at a single site. Fig. 2 shows the path diagram for a single site with three collocated samplers. This is the simplest case in which it is possible for a calibration model to simultaneously estimate both systematic error (bias) and random error (imprecision) without making arbitrary assumptions. Because the true $PM_{2.5}$ mass concentrations (μ) are unknown, measurements from the first device (X_1) are arbitrarily used as the reference so that β_1 is constrained to equal one, α_1 (not shown in Fig. 2) is constrained to equal zero, and the biases for samplers 2 and 3 are then estimated relative to sampler 1. The imprecision standard deviations (σ_i) for each sampler are also estimated by the model. In this study, as in our previous study, we arbitrarily use the FRM as the reference sampler. (The relative biases and differences in imprecision among the various models of FRM samplers are assumed to be negligible).This basic design is then used as a building block to handle a more complex analysis. For example, in our previous work (Bilonick et al., 2015), the classic model was extended, using stratification, to account for the nonlinear effect of temperature on $PM_{2.5}$ mass concentration measurements made using a 50 °C TEOM monitor. Here we extend the model further to also include samplers located at multiple monitoring sites.

3.2. Structural equation model

Each path diagram represents a structural equation model (SEM), which is a set of simultaneous equations that include one or more latent variables. As shown in more detail in our previous study (Bilonick et al., 2015), the relationship between any pair of devices can be derived. By taking expectations (averaging out the random error using expected values), the calibration equations can be determined. The calibration equations corresponding to the path diagram in Fig. 2 are:

$$1:2\ \ E[X_1]=\left[\alpha_1-\alpha_2\frac{\beta_1}{\beta_2}\right]+\frac{\beta_1}{\beta_2}E[X_2],\ \ E[X_2]=\left[\alpha_2-\alpha_1\frac{\beta_2}{\beta_1}\right]+\frac{\beta_2}{\beta_1}E[X_1]$$

$$1:3\ \ E[X_1]=\left[\alpha_1-\alpha_3\frac{\beta_1}{\beta_3}\right]+\frac{\beta_1}{\beta_3}E[X_3],\ \ E[X_3]=\left[\alpha_3-\alpha_1\frac{\beta_3}{\beta_1}\right]+\frac{\beta_3}{\beta_1}E[X_1]$$

$$2:3\ \ E[X_2]=\left[\alpha_2-\alpha_3\frac{\beta_2}{\beta_3}\right]+\frac{\beta_2}{\beta_3}E[X_3],\ \ E[X_3]=\left[\alpha_3-\alpha_2\frac{\beta_3}{\beta_2}\right]+\frac{\beta_3}{\beta_2}E[X_2]$$

Table 2
Total numbers of 24-h averaged $PM_{2.5}$ mass concentration measurements by site, sampler type, and collocation status. The first two digits of the site label indicate the state in which the site was located, as follows: 24 = Maryland, 39 = Ohio, 42 = Pennsylvania, and 54 = West Virginia. The *Collocation* column indicates which samplers were collocated. For example, for site *39-121-9000*, samplers of types 9 and 12 were collocated and two samplers of type 12 were collocated. This does not necessarily mean that all three samplers operated on the same days, but there was overlap so that a separate calibration model could be fitted. For site *42-125-5001*, samplers of type 1 and 10 were collocated and, separately, samplers of type 4 and 10 were collocated. For site *39-99-5*, two samplers of type 2 both operated on some days.

Site		Latitude	Longitude	Collocation	Sampler type															Total
Label	ID				1	2	3	4	5	6	7	8	9	10	12	14	16	17	19	
24-23-2	8	39.70595	−79.01200	9,17	0	0	0	0	0	0	0	0	553	0	0	0	0	1384	0	1937
39-121-9000	19	39.94283	−81.33761	9,12,12	0	0	0	0	0	0	0	0	913	0	177	0	0	0	0	10909
39-155-7	10	41.21417	−80.78750	2,14	0	2483	0	0	0	0	0	0	0	0	0	1130	0	0	0	3613
39-67-HOP	42	40.32000	−80.90000		0	586	0	0	0	0	0	0	0	0	0	0	0	0	0	586
39-81-1001	30	40.32194	−80.60639	2,2,10	0	1908	0	0	0	0	0	0	0	92	0	0	0	0	0	200
39-81-16	23	40.36278	−80.61556		0	536	0	0	0	0	0	0	0	0	0	0	0	0	0	536
39-81-17	46	40.36610	−80.61500	2,10	0	547	0	0	0	0	0	0	0	125	0	0	0	0	0	672
39-81-FRA	6	40.38000	−80.62000	2,14	0	640	0	0	0	0	0	0	0	0	0	587	0	0	0	1227
39-99-14	7	41.09587	−80.65843	2,10,14	0	1491	0	0	0	0	0	0	0	391	0	2277	0	0	0	4159
39-99-5	13	41.11111	−80.64528	2,2	0	2366	0	0	0	0	0	0	0	0	0	0	0	0	0	2366
42-125-200	22	40.17056	−80.26139		1112	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1112
42-125-5	12	40.14667	−79.90222	1,1,17	1340	0	0	0	0	0	0	0	0	0	0	0	0	809	0	2149
42-125-5001	21	40.44528	−80.42083	1,10; 4,10	1510	0	0	1633	0	0	0	0	0	449	0	0	0	0	0	3592
42-129-8	31	40.30469	−79.50567	1,10; 4,10	512	0	0	582	0	0	0	0	0	466	0	0	0	0	0	1560
42-129-SVC	40	40.29000	−79.40000		0	674	0	0	0	0	0	0	0	0	0	0	0	0	0	647
42-21-11	9	40.30972	−78.91500	1,17	1108	0	0	0	0	0	0	0	0	0	0	0	0	1510	0	2618
42-3-1008	32	40.61861	−79.72722		1037	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1037
42-3-116	27	40.47361	−80.07722		718	0	0	0	0	0	0	0	0	0	0	0	0	0	0	718
42-3-1301	25	40.40250	−79.86028	1,1	1246	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1246
42-3-131	29	40.28944	−80.00500		180	0	0	0	0	0	0	0	0	0	0	0	0	0	0	180
42-3-133	45	40.26013	−79.88650		156	0	0	0	0	0	0	0	0	0	0	0	0	0	0	156
42-3-21	33	40.41361	−79.94139	1,10	732	0	0	0	0	0	0	0	0	143	0	0	0	0	0	875
42-3-3007	44	40.29444	−79.88667		455	0	0	0	0	0	0	0	0	0	0	0	0	0	0	455
42-3-64	4	40.32361	−79.86833	1,1,10,14	3562	0	0	0	0	0	0	0	0	285	0	3210	0	0	0	7057
42-3-67	36	40.38194	−80.18556		971	0	0	0	0	0	0	0	0	0	0	0	0	0	0	971
42-3-8	5	40.46556	−79.96111	1,1,5,6,9,10,14	3355	0	0	0	129	439	0	0	122	767	0	3057	0	0	0	7869
42-3-9002	24	40.54694	−79.78389		369	0	0	0	0	0	0	0	0	0	0	0	0	0	0	369
42-3-93	35	40.60722	−80.02083		412	0	0	0	0	0	0	0	0	0	0	0	0	0	0	412
42-3-95	26	40.48694	−80.18806		433	0	0	0	0	0	0	0	0	0	0	0	0	0	0	433
42-3-97	28	40.55306	−80.20333		56	0	0	0	0	0	0	0	0	0	0	0	0	0	0	56
42-3-BRU	2	40.30650	−79.97940	1,14,16	1597	0	0	0	0	0	0	0	0	0	0	1056	436	0	0	3089
42-3-SCH	1	40.43950	−79.94050	7,8,16	0	0	0	0	0	0	319	348	0	0	0	0	378	0	0	1045
42-5-1	3	40.81418	−79.56475		0	0	0	0	0	0	0	0	0	0	0	0	2727	0	0	2727
42-59-2	34	39.80933	−80.26567	5,6	0	0	0	0	129	286	0	0	0	0	0	0	0	0	0	415
42-7-14	37	40.74780	−80.31644	1,1	1212	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1212
42-85-100	39	41.21501	−80.48478		1021	0	0	1657	0	0	0	0	0	0	0	0	0	0	0	2678
42-85-9000	20	41.42691	−80.14507	9,12	0	0	0	0	0	0	0	0	924	0	143	0	0	0	0	1067
54-29-1004	14	40.42154	−80.58072		740	0	0	392	0	0	0	0	0	0	0	0	0	0	0	1132
54-29-TOM	41	40.54000	−80.58000		0	662	0	0	0	0	0	0	0	0	0	0	0	0	0	662
54-49-6	38	39.48148	−80.13467		661	0	371	0	0	0	0	0	0	0	0	0	0	0	0	1032
54-51-1002	11	39.91596	−80.73386	1,10,19	784	0	0	860	0	0	0	0	0	264	0	0	0	0	1031	2939
54-61-3	15	39.64937	−79.92087	1,6	869	0	295	14	0	121	0	0	0	0	0	0	0	0	0	1299
54-69-10	47	40.11488	−80.70097		78	0	0	400	0	0	0	0	0	0	0	0	0	0	0	478
54-69-8	16	40.06383	−80.72050		687	0	0	0	0	0	0	0	0	0	0	0	0	0	0	687
54-69-WJU	43	40.07000	−80.69000		0	416	0	0	0	0	0	0	0	0	0	0	0	0	0	416
54-9-11	17	40.39458	−80.61202	1,1; 4,4	1532	0	0	632	0	0	0	0	0	0	0	0	0	0	0	2164
54-9-5	18	40.34102	−80.59664		760	0	0	400	0	0	0	0	0	0	0	0	0	0	0	1160
Total					29205	12309	666	6570	258	846	319	348	2512	2982	320	11317	3541	3703	1031	75927

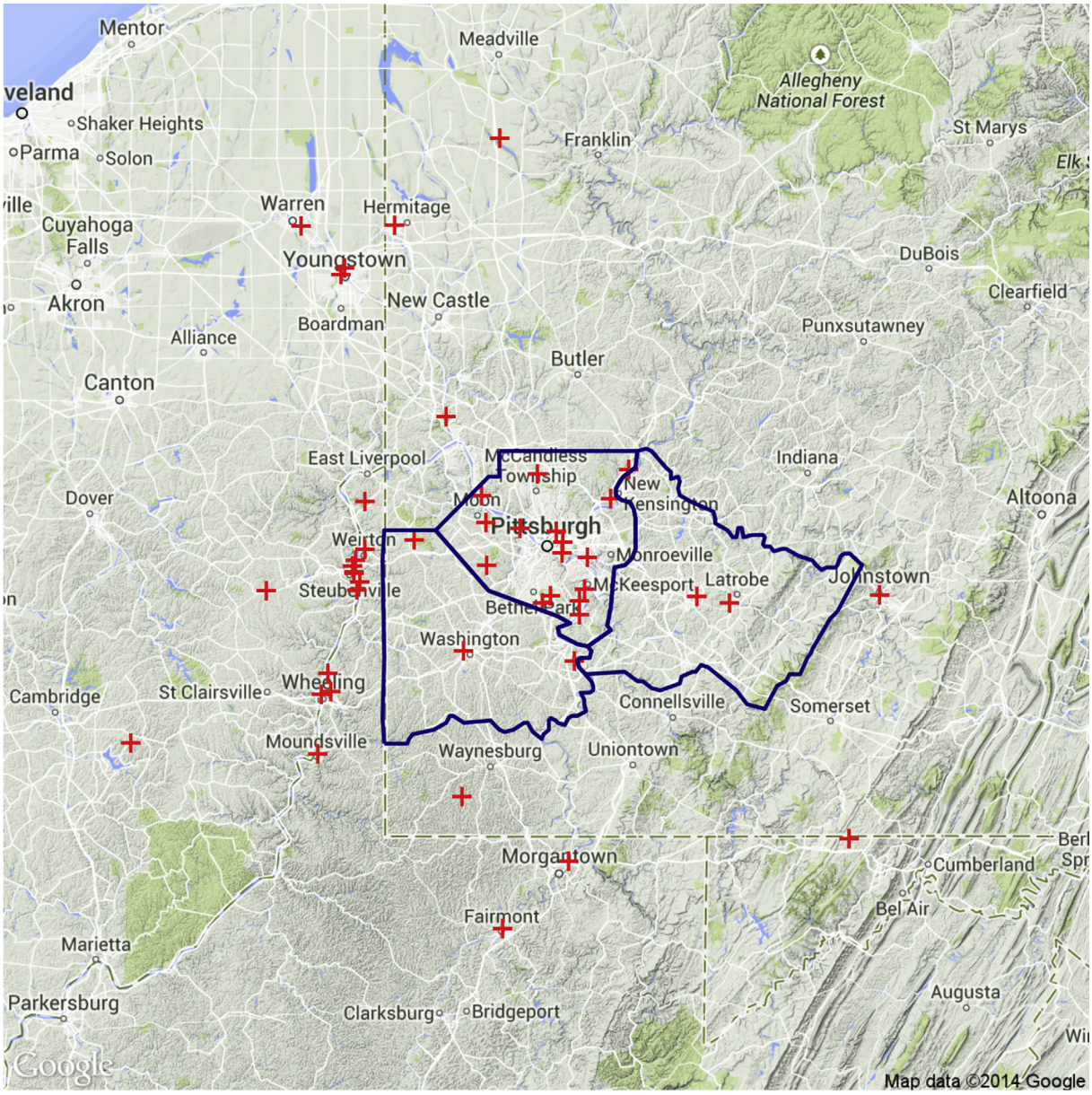


Fig. 1. Locations of the 47 monitoring sites (+) in Pennsylvania, Ohio, West Virginia, and Maryland. Boundaries of the counties constituting the three-county study area are outlined.

The calibration intercepts and slopes are invariant to the choice of constraints in the SEM. The observed covariance matrix **S** is:

$$\begin{pmatrix} S_1^2 & S_{12} & S_{13} \\ & S_2^2 & S_{23} \\ & & S_3^2 \end{pmatrix}$$

where S_i^2 is the variance of the measurements X_i, and S_{ij} is the covariance between measurements X_i and X_j.

A simple approach to providing estimates of the model parameters is to use the method of moments. In the method of moments, the theoretical covariance matrix of the measurements is derived from the model and equated to the observed covariance matrix. The theoretical covariance matrix is a function of the model parameters β_2, β_3, σ_1, σ_2, σ_3, and σ, and the solution of the simultaneous equations provides the parameter estimates. (If the sample means are included, α_2, α_3, and $\overline{\mu}$ can also be estimated.) With three different types of measurements in the simple one-factor model shown in Fig. 2, there are as many equations as unknown parameters, and thus a unique solution results. To illustrate, first note that the theoretical covariance matrix Σ for this model is:

$$\Sigma = \mathbf{B}\sigma^2\mathbf{B}' + \mathbf{E} \tag{1}$$

where $B = [\beta_1 \quad \beta_2 \quad \beta_3]$ and **E** denotes the 3×3 residual variance matrix:

$$\begin{pmatrix} \sigma_1^2 & 0 & 0 \\ & \sigma_2^2 & 0 \\ & & \sigma_3^2 \end{pmatrix}$$

In this case, Σ is given by:

$$\begin{pmatrix} \beta_1^2\sigma^2 + \sigma_1^2 & \beta_1\beta_2\sigma^2 & \beta_1\beta_3\sigma^2 \\ & \beta_2^2\sigma^2 + \sigma_2^2 & \beta_2\beta_3\sigma^2 \\ & & \beta_3^2\sigma^2 + \sigma_3^2 \end{pmatrix}$$

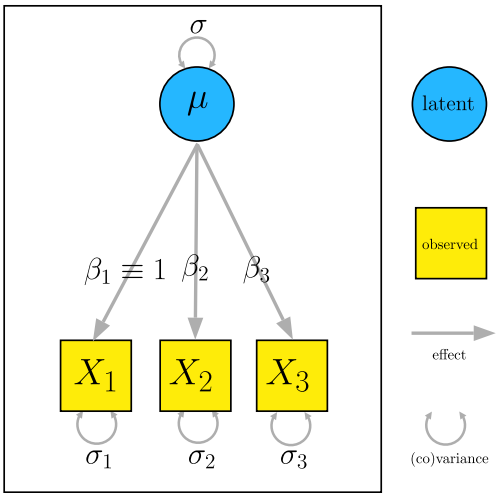


Fig. 2. Path diagram for the measurement error model describing three collocated $PM_{2.5}$ samplers. Here μ denotes the square-root transformed true but unobserved 24-h averaged $PM_{2.5}$ mass concentrations and is assumed to be normally distributed with mean $\bar{\mu}$ and standard deviation σ. Each sampler provides a measurement, X_1, X_2, and X_3. The single-ended arrows indicate the direction of influence - each X_i is a linear function of μ with intercept α_i (not shown) and scale (slope) β_i. The curved, double-ended arrows denote standard deviations. The residual standard deviation for the ith sampler is σ_i and denotes the imprecision. Not including intercepts, there are six parameters to be estimated: β_2, β_3, σ_1, σ_2, σ_3, and σ.

The method of moments equates the observed covariance matrix **S** to the theoretical covariance matrix **Σ** resulting in six equations (setting $\beta_1 = 1$) in six unknowns:

$$S_1^2 = \sigma^2 + \sigma_1^2$$
$$S_2^2 = \beta_2^2\sigma^2 + \sigma_2^2$$
$$S_3^2 = \beta_3^2\sigma^2 + \sigma_3^2$$
$$S_{12} = \beta_2\sigma^2$$
$$S_{13} = \beta_3\sigma^2$$
$$S_{23} = \beta_2\beta_3\sigma^2$$

This easily leads to the solution for the method of moments parameter estimates:

$$\beta_2 = S_{23}/S_{13}$$
$$\beta_3 = S_{23}/S_{12}$$
$$\sigma_1^2 = S_1^2 - \frac{S_{12}S_{13}}{S_{23}}$$
$$\sigma_2^2 = S_2^2 - \frac{S_{12}S_{23}}{S_{13}}$$
$$\sigma_3^2 = S_3^2 - \frac{S_{13}S_{23}}{S_{12}}$$
$$\sigma^2 = \frac{S_{12}S_{13}}{S_{23}}$$

The solution depends on the constraint used to ensure a unique solution (i.e., identifiability) and will change, for example, if β_2, instead of β_1, is set to 1. Other constraints are possible. If the product $\beta_1\beta_2\beta_3$ is constrained to 1 (effectively constraining the geometric average of the βs), the estimate for σ^2 will be $\sqrt[3]{S_{12}S_{13}S_{23}}$ and the estimates for the βs will also differ from the current solution. The ratio of the βs and the ratio of the imprecision variances are, however, invariant to the choice of these constraints.

In practice, the method of maximum likelihood (Lehmann and Casella, 1998) is preferred to the method of moments for both simple and complex models because the method of maximum likelihood usually provides superior estimates. If there are no missing values, the model parameter estimates are easily computed using the method of maximum likelihood. Otherwise, full information maximum likelihood accommodates missing values.

3.3. Linked sub-models for multiple sites

Sites *42-3-8* (ID = 5) and *42-85-9000* (ID = 20) will be used to illustrate linking sub-models and stratification by temperature. Site ID = 20 had only two collocated samplers of type 9 (Met One SASS speciation) and type 12 (CASTNet speciation). Site ID = 5 had six samplers, but for this illustration we will only consider the type 1 (R & P Partisol-Plus Model 2025 Sequential Air Sampler FRM), type 9 (IMPROVE), and type 14 (Rupprecht & Patashnick 50 °C TEOM) samplers. The first step in developing a comprehensive model to calibrate all measurement devices that are spread across multiple sites is to construct individual models for each site. The top leftmost path diagram shown in Fig. 3 illustrates the measurement error model for Site ID = 5 for the lowest temperature stratum. The circle labeled with μ represents the unknown true $PM_{2.5}$ concentrations where the first subscript denotes the site ID and the second subscript describes the temperature stratum. The double-ended curved arrow pointing at the circle denotes the standard deviation of the true values labeled as σ_{5-1}. There are three single-arrow paths pointing to the three collocated measurements each denoted by an X in a square box. The third subscript for each X denotes the sampler type. The path labels are the systematic bias β parameters with the FRM sampler used as a reference by constraining its coefficient to equal 1. The imprecision standard deviations for each sampler are denoted by the σs labeling the double-ended curved arrows pointing to each yellow square box. The intercept parameters (mean of the true values and the α intercepts) are not shown for simplification. With just these three samplers at this site for this temperature stratum, there is sufficient information to fully estimate all of the model parameters. Site ID = 20 for the first temperature stratum is shown as the leftmost path diagram on the bottom row. At this site there are only two samplers so that by itself, there would not be sufficient information to fully estimate all of the model parameters without additional constraints. The constraint used to *link* the two sub-models is to equate the β parameters for the type 9 samplers at each site. This is accomplished simply by using the same label $\beta_{\#-\#-9}$ for both samplers. The first # indicates that the label is independent of the site. This approach allows sampler type 9 to be calibrated against the FRM at site ID = 5, and sampler type 12 at site ID = 20 can then be calibrated indirectly against the FRM through sampler type 9. This represents the assumption that the two samplers of type 9 at each site operate in the same manner.

3.4. Temperature stratification needed for TEOM samplers

The final step in model construction is to extend the linking across all temperature strata as shown in Fig. 3. In each path diagram, the second subscript denoting the temperature stratum changes from column to column for μ, the measurements X and the standard deviation parameters. For the TEOM sampler (type = 14), the bias parameter label also changes with temperature. For samplers that do not depend on temperature (samplers of type 1, 9 and 12), the β labels are the same across all temperature strata, i.e., the second subscript is always a #. The method used to define the eight temperature strata was described in Bilonick et al. (2015).

3.5. Linked sites

As previously discussed, some sites with collocated samplers of

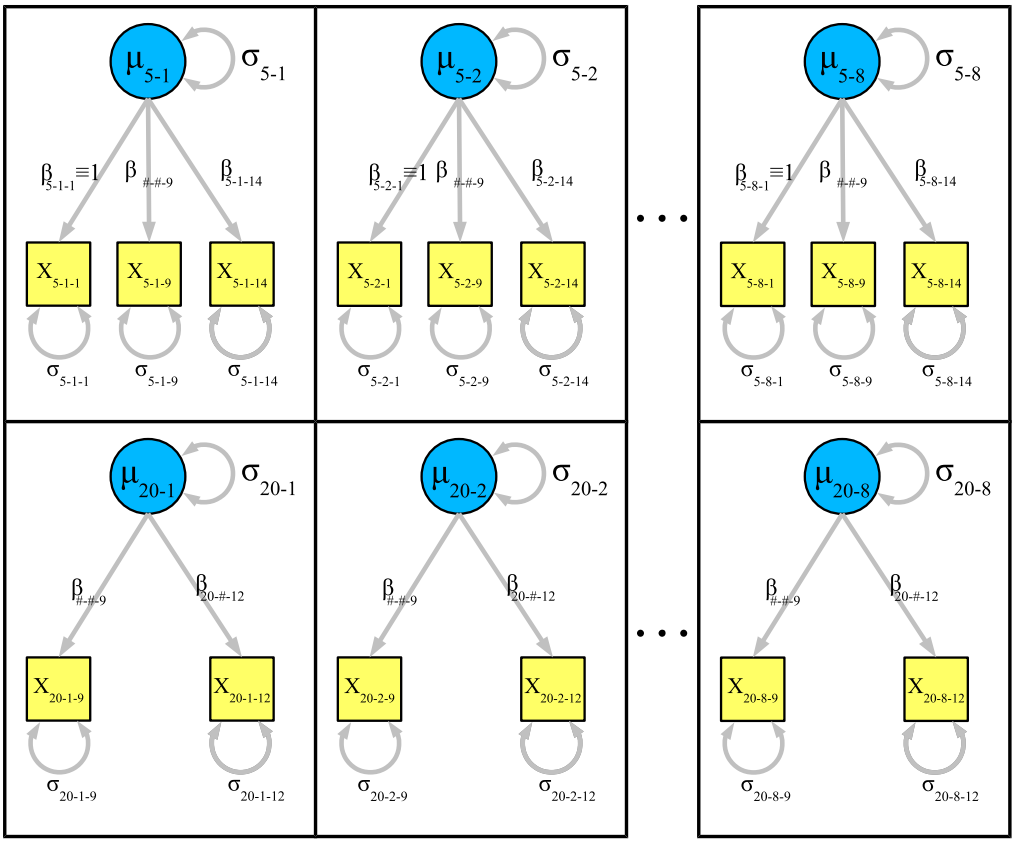


Fig. 3. Simplified illustration of temperature stratification for linked set 3 with eight strata (1, 2, … 8) across two sites (5 and 20) in order to determine the dependency of TEOM sampler $PM_{2.5}$ mass concentration measurements on temperature relative to the Federal Reference Method samplers. Only three sampler types are illustrated at site 5 (sampler types 1, 9, and 14). Note that μ_{5-1} denotes the true values at site 5 for the first (lowest) temperature stratum (stratum 1). X_{5-1-9} denotes the measured $PM_{2.5}$ concentration at site 5, first temperature stratum, and sampler type 9. By using the same label $\beta_{\#-\#-9}$ for all temperature strata for site 5, the coefficient is assumed not to depend on temperature. By using the same label at site 20, this provided a reference for sampler 12 at this site. Otherwise, there would be no way to calibrate sampler 12 because there were only two collocated samplers at this site. Conversely, the β coefficients for sampler type 14 (TEOM 50 °C) have different labels for each temperature stratum for site 5 in order to determine how the relationship between the FRM and TEOM changes with temperature.

different types had only two samplers and/or lacked an FRM. Thus, in order to produce calibration equations relating all of the non-FRM samplers to the FRM sampler, it was necessary to construct subsets of sites as shown in Table 3. The seven linked sets shown in this table cover all of the types of samplers and all of the sites with collocated samplers, except for sites that only had collocated FRM samplers for which calibration was not needed. Fortunately, there were no sites in this study that had only one non-FRM sampler.

4. Measurement error model fitting

The R statistical language and environment (Version 2.13.1) (R Core Team, 2013) was used for statistical analysis and graphics. The OpenMx R package (Version 1.0.6–1581) (Boker et al., 2011, 2012) was used to estimate the SEM parameters using full information maximum likelihood. Modeling of the temperature relationship using nonlinear regression was performed using the `nls` R function, and the nlme R package (Version 3.1–11) was used to fit nonlinear mixed effects models (Pinheiro et al., 2013). It was necessary to square root transform the $PM_{2.5}$ concentrations to ensure approximate normality.

5. Non-TEOM sampler results

5.1. Parameter estimates

In our initial modeling, we determined that the TEOM was the only sampler type for which bias (relative to the FRM sampler) tended to show a dependency with temperature. Therefore, the final models used assumed that the bias parameters for all non-TEOM samplers were not temperature-dependent, and only bias parameters for TEOMs were allowed to vary with temperature. The parameter estimates for the non-TEOM samplers are shown in Table 4. Except for the single dichotomous sampler (type 7), each sampler type was used at multiple sites. Summary measures for multiple sites were computed as the simple arithmetic mean for α, as the geometric mean for β, and as the root mean squared for σ_i/β_i. For β, the range of estimates for each sampler type tends to be relatively narrow. The one exception occurs for the Met One SASS speciation sampler (type 10) at site 46 (*39-81-17*), which has an unusually low estimate for β. The confidence intervals for β in general were reasonably narrow indicating reasonably precise estimates.

For the Desert Research Institute Sequential Filter Sampler (type 6), the estimated β is below 1 for all three sites and is statistically significantly below 1 for two of the three sites. For the dichotomous sampler (type 7), the estimated β is very close to 1 and not statistically significantly different from 1. For the IMPROVE speciation sampler (type 9), the β at site 5 is slightly above 1 and the β at site 8 is slightly below 1, and neither is statistically significantly different than 1. For the Met One SASS speciation sampler (type 10), eight of the nine sites had βs very near 1 although four had narrow enough 95% confidence intervals to be statistically significantly different than 1 (three of these were above 1 and one was below 1). As noted

Table 3
Sets of linked monitoring sites. In order to be able to calibrate all sampler types, it was necessary to group individual sites with collocated samplers into seven sets according to available overlap in the sampler types. FRM samplers (1–5 and 8) are in **bold** and whenever possible were used to link sites. If FRMs were not available, the linked samplers are underlined. Duplicated samplers of the same type are in *italics*.

Set	Site ID	Sampler types						FRM imprecision SD
1	1	7	**8**	16				0.201
	2	**1**	14	16				
	33	**1**	10					
2	4	***1***	10	14				0.100
	9	**1**	17					
	31	**1**	**4**	10				
3	5	***1***	**5**	6	9	10	14	0.103
	19	9	*12*					
	20	9	12					
4	6	**2**	14					0.216
	21	**1**	**4**	10				
	30	***2***	10					
	46	**2**	10					
5	7	**2**	10	14				0.148
	10	**2**	14					
6	11	**1**	**4**	10	19			0.117
	15	**1**	6					
	34	**5**	6					
7	8	9	17					0.155
	12	***1***	17					
Root Mean Sq.								0.155

above, the estimate for β at site 46 was substantially lower than 1 and was statistically significantly different than 1 with an upper bound of just 0.866. For the CASTNET speciation sampler (type 12), both estimates for β were above 1, with one site having a statistically significant difference from 1. Finally, for the Beta Attenuation Monitor (type 17), the β for one site was somewhat above 1 and the β for the other was somewhat below 1, with both statistically significantly different from 1.

In general, the bias parameter α, while needed for calibration, is difficult to interpret. The exception occurs when the corresponding parameter β equals (or at least is near) 1. When $\beta = 1$, the parameter α represents the constant bias between the sampler and the FRM sampler. For example, the Dichotomous sampler at site 1 and the Met One SASS sampler at site 7 have estimated βs very nearly equal to 1. Thus, these samplers at these sites had a nearly constant bias relative to the FRM sampler. When β differs from 1, the bias changes with the true $PM_{2.5}$ concentration. Calibration is discussed in more detail in the next section.

The typical value for the FRM imprecision standard deviation (σ_i) is given in Table 3 as 0.155 $(\mu g\ m^{-3})^{0.5}$. Table 4 shows that non-FRM/non-TEOM samplers tended to have higher scale-adjusted imprecision standard deviations (σ_i/β_i). The exception was the IMPROVE sampler, which had a scale-adjusted imprecision standard deviation that was similar to the FRM samplers.

5.2. Calibration results

The calibration parameter estimates shown in Table 4 can be used to write the calibration equation relating each sampler at a site to the FRM sampler. For example, for the CASTNET speciation sampler (type 12) at site 20:

$$E[X_{20-CASTNet-\#}] = \left(-0.270 + 1.163\sqrt{E[X_{20-FRM-\#}]}\right)^2 \quad (2)$$

Equivalently, to predict the FRM equivalent value from the observed speciation sampler measurement:

$$E[X_{20-FRM-\#}] = \left(\frac{0.270}{1.163} + \frac{1}{1.163}\sqrt{E[X_{20-CASTNet-\#}]}\right)^2 \quad (3)$$

In a similar fashion, a calibration equation is available for each non-TEOM sampler at each site to convert the observed measurement to an FRM-equivalent value. Although a site-specific calibration equation is available for all of the non-TEOM samplers in this study, had there been a site without collocated measurements, the parameter estimates developed for the same sampler type at a

Table 4
Calibration parameter estimates on the square root transformed scale for speciation, dichotomous, and beta attenuation monitor samplers. For β, the mean is the geometric mean. For α, the mean is the arithmetic mean. For the scale-adjusted imprecision standard deviation (σ_i/β_i), the mean is the root mean square.

Sampler		Site ID	β			α			σ_i/β_i
Description	Type		Estimate	Lower	Upper	Estimate	Lower	Upper	
DRI SFS	6	5	0.889	0.846	0.932	0.249	0.078	0.421	0.503
		15	0.877	0.759	0.994	0.689	0.233	1.143	0.675
		34	0.927	0.817	1.036	0.575	0.182	0.973	0.759
		Mean	0.897			0.504			0.654
Dichotomous	7	1	1.003	0.965	1.042	−0.139	−0.294	0.017	0.318
IMPROVE	9	5	1.016	0.977	1.055	−0.087	−0.239	0.065	0.191
		8	0.951	0.889	1.017	−0.165	−0.404	0.060	0.105
		Mean	0.982			−0.126			0.154
Met One SASS	10	4	0.977	0.946	1.008	0.328	0.192	0.464	0.373
		5	0.983	0.968	0.999	0.243	0.182	0.304	0.201
		7	0.997	0.972	1.023	0.127	0.029	0.226	0.177
		11	0.954	0.923	0.985	0.273	0.153	0.392	0.212
		21	1.050	1.014	1.087	−0.024	−0.158	0.105	0.211
		30	0.935	0.835	1.034	0.697	0.304	1.092	0.491
		31	0.968	0.944	0.992	0.231	0.140	0.321	0.222
		33	1.039	0.975	1.103	0.006	−0.262	0.270	0.254
		46	0.785	0.705	0.866	0.556	0.214	0.894	0.701
		Mean	0.962			0.271			0.357
CASTNet	12	19	1.100	0.962	1.261	−0.142	−0.686	0.328	0.277
		20	1.163	1.030	1.320	−0.270	−0.765	0.170	0.086
		Mean	1.131			−0.206			0.205
BAM	17	9	1.078	1.053	1.104	−0.220	−0.319	−0.121	0.264
		8/12	0.949	0.902	0.997	0.126	−0.055	0.305	0.463
		Mean	1.012			−0.047			0.377

nearby site, or the mean parameter estimates for that sampler type, could be applied to best approximate the appropriate calibration equations.

6. TEOM sampler results

6.1. Modeling the sigmoidal temperature effect using nonlinear regression

As shown in Table 3, there were 9 TEOM samplers of three types (14–50 °C, 16–30 °C, and 19 - FDMS) at 8 different sites. Using the linked subsets and the temperature stratification strategy described in the path diagram of Fig. 3, calibration model parameter estimates were computed as a function of temperature. Eight temperature strata covering temperatures ranging from about −5 to 25 °C were used as described in Bilonick et al. (2015). The parameter estimates for the 50 °C (type 14) TEOMs are shown in Table 5, for the 30 °C (type 16) TEOMs in Table 6, and for the FDMS (type 19) TEOM in Table 7. The parameter estimates for both α and β for many of the TEOMs show a nonlinear pattern.

To describe the general relationship, nonlinear regression was used to fit a sigmoidal (logistic) model using the R `SSfpl` self-starting four-parameter logistic function with the `nls` nonlinear regression function. The four-parameter sigmoidal model is described by the following equation:

$$y(T) = A + \frac{B - A}{1 + e^{\frac{T_{mid} - T}{S}}} + \varepsilon \tag{4}$$

where $y(T)$ denotes a particular bias parameter (either α or β) and T denotes the temperature in °C. This model was fitted to the eight observed temperature stratum means and the corrresponding values for α and for β, respectively. A denotes the fixed effect asymptote as $T \to -\infty$, B denotes the fixed effect asymptote as $T \to +\infty$, T_{mid} is a fixed effect measure of location (the temperature for the single inflection point of the logistic curve), and S is a fixed effect measure of spread (the distance between T_{mid} and the temperature at which y reaches about 3/4 of the distance between B and A). (When y increases as T increases, $B - A$ is positive, and when y decreases as T increases, $B - A$ is negative.)

Fig. 4 plots the estimated parameters for each temperature stratum with the corresponding sigmoidal model (or other appropriate model) superimposed. For β, a clear sigmoidal pattern was observed for all samplers with the exception of the 30 °C TEOM sampler (type 16) at site 2 and the 50 °C TEOM sampler (type 14) at site 6. Otherwise, the pattern was the same - β was below 1 for colder temperatures and approached 1 as the temperature increased - except for the FDMS TEOM (type 19) at site 11 which showed the opposite pattern (β was near 1 for colder temperatures and quickly dropped below 1 as the temperature increased). This opposite behavior might be expected, as the FDMS TEOM is specifically designed to avoid the loss of semi-volatile material that affects the 50 °C TEOM at low temperatures. Several other studies (Schwab et al., 2006; Grover et al., 2005) have observed FDMS TEOM samplers to be biased high relative to collocated FRM samplers and have attributed this bias to partial loss of semivolatile species (e.g., ammonium nitrate and semivolatile organic material) from the FRM measurement. The single FDMS TEOM sampler used in PARIES did not exhibit this behavior; rather it was statistically equivalent to the FRM sampler at low temperatures and was biased low relative to the FRM at higher temperatures (roughly 10 °C and above). The cause of the discrepancy between our study and these previous studies is unknown; we suspect it could result from differences in $PM_{2.5}$ chemical composition among the studies and, in particular, from differences in the measurement of particle-bound water, which has been shown to contribute substantially to the mass of $PM_{2.5}$ determined using the FRM in the Pittsburgh region during summer (Rees et al., 2004).

For α, all samplers except for the type 16 sampler at site 2 showed a sigmoidal pattern. Not surprisingly, the FDMS TEOM (type 19) at site 11 showed a pattern that was opposite of the general pattern. For the cases indicated above in which sigmoidal patterns were not observed, simple linear regression models were fitted instead.

Table 8 shows the estimated calibration model parameters for all nine TEOMs. For example, the calibration between the 50 °C TEOM and the FRM sampler at site 4 is given by:

$$E[TEOM(T)] = \left(\alpha(T) + \beta(T)\sqrt{E[FRM]}\right)^2 \tag{5}$$

$$E[FRM(T)] = \left(-\frac{\alpha(T)}{\beta(T)} + \frac{1}{\beta(T)}\sqrt{E[TEOM]}\right)^2 \tag{6}$$

where

$$\alpha(T) = 0.333 + \frac{-0.152 - 0.333}{1 + e^{\frac{18.524 - T}{0.951}}} \tag{7}$$

and

$$\beta(T) = 0.796 + \frac{1.082 - 0.796}{1 + e^{\frac{16.542 - T}{4.578}}} \tag{8}$$

6.2. Nonlinear mixed effects model accounting for temperature

Four of the 50 °C (type 14) TEOMs had similar, well-defined sigmoidal patterns for α and for β. For these samplers at sites 4, 5, 7, and 10, a nonlinear mixed effects model was fitted uing the `nlme` and `SSfpl` functions from the nlme R package. The nonlinear sigmoidal mixed effects model is similar to the model shown by equation (4) and is described as follows:

$$y(T) = (A + a) + \frac{(B + b) - (A + a)}{1 + e^{\frac{(T_{mid} + t) - T}{S + s}}} + \varepsilon \tag{9}$$

where a, b, t, and s are the corresponding random effects (site-specific effects), assumed to be normally distributed with means of zero and standard deviations σ_a, σ_b, σ_t, and σ_s, respectively. When fitting these models for α and β, the reciprocal of the confidence interval width was used as the weight for each calibration parameter estimate so that less precise estimates were given less weight. Also, a variance function was used to allow each site to have its own variance for ε using site 4 as the (arbitrary) reference. Random effects for α were included for A (a) and for T (t). For β, random effects were included for A (a) and S (s).

Tables 9 and 10 show the estimated parameters for the two logistic non-linear mixed effects calibration models. In general, the confidence intervals for A and B for both α and β are reasonably narrow. The remaining parameters tend to have fairly wide confidence intervals. The fixed effects parameters (A B, T_{mid}, and S) describe the *typical* temperature effect. Although all of the 50 °C (type 14) TEOMs can be directly calibrated to the FRM sampler in this study, this relationship could be used to calibrate a 50 °C TEOM located in the same region that had no available collocated FRM

Table 5
Calibration parameter estimates and 95% confidence intervals on the square root scale for TEOM 50 °C samplers (type 14) as a function of temperature.

Site ID	°C	β			α			σ_i/β_i
		Estimate	Lower	Upper	Estimate	Lower	Upper	
2	−6.026	0.585	0.473	0.690	0.965	0.594	1.361	0.828
	0.547	0.595	0.507	0.681	1.010	0.705	1.325	0.867
	5.344	0.730	0.601	0.855	0.686	0.279	1.105	0.838
	10.016	0.841	0.707	0.971	0.464	0.027	0.911	0.627
	14.554	0.871	0.730	1.006	0.435	−0.072	0.959	0.747
	17.999	0.909	0.752	1.058	0.308	−0.273	0.917	0.722
	21.114	0.848	0.707	0.984	0.499	−0.109	1.135	0.735
	24.333	0.827	0.708	0.946	0.569	−0.069	1.204	0.685
4	−5.834	0.801	0.768	0.834	0.220	0.094	0.346	0.416
	0.666	0.783	0.754	0.812	0.401	0.286	0.516	0.463
	5.360	0.841	0.817	0.866	0.303	0.200	0.405	0.427
	10.018	0.856	0.834	0.878	0.387	0.294	0.480	0.376
	14.355	0.890	0.869	0.890	0.352	0.261	0.443	0.326
	18.054	0.960	0.935	0.984	0.147	0.034	0.259	0.346
	21.127	1.026	1.003	1.049	−0.115	−0.230	−0.000	0.274
	24.396	1.028	1.003	1.053	−0.156	−0.299	−0.014	0.285
5	−5.829	0.700	0.627	0.774	0.454	0.190	0.717	0.718
	0.671	0.722	0.672	0.772	0.519	0.340	0.698	0.565
	5.345	0.777	0.724	0.830	0.458	0.277	0.638	0.541
	10.013	0.860	0.793	0.927	0.311	0.092	0.530	0.542
	14.353	0.942	0.888	0.996	0.119	−0.073	0.310	0.446
	18.060	1.011	0.960	1.062	−0.047	−0.244	0.149	0.416
	21.146	1.050	1.006	1.093	−0.282	−0.472	−0.093	0.375
	24.394	1.020	0.987	1.053	−0.180	−0.351	−0.009	0.302
6	−5.644	0.937	0.812	1.072	−0.782	−1.251	−0.344	0.224
	0.681	0.971	0.905	1.039	−0.790	−1.046	−0.540	0.193
	5.202	0.996	0.907	1.086	−0.873	−1.238	−0.513	0.349
	10.228	0.996	0.934	1.058	−0.718	−0.960	−0.478	0.221
	14.642	0.983	0.901	1.065	−0.580	−0.951	−0.211	0.382
	17.784	0.915	0.830	1.001	−0.194	−0.559	0.166	0.273
	21.080	1.006	0.863	1.150	−0.500	−1.198	0.194	0.521
	24.197	1.048	0.941	1.155	−0.720	−1.282	−0.165	0.253
7	−5.875	0.706	0.647	0.765	0.307	0.093	0.521	0.509
	0.706	0.725	0.671	0.780	0.420	0.219	0.620	0.487
	5.364	0.761	0.694	0.829	0.358	0.127	0.591	0.548
	9.974	0.879	0.806	0.952	0.134	−0.113	0.378	0.448
	14.300	1.012	0.956	1.068	−0.214	−0.412	−0.016	0.269
	18.122	1.027	0.982	1.072	−0.242	−0.414	−0.073	0.223
	21.141	1.056	1.015	1.097	−0.423	−0.594	−0.254	0.200
	24.340	1.034	0.995	1.072	−0.303	−0.488	−0.120	0.030
10	−5.801	0.581	0.523	0.642	1.039	0.810	1.257	0.299
	0.680	0.674	0.611	0.735	0.821	0.590	1.052	0.260
	5.361	0.741	0.676	0.807	0.742	0.529	0.953	0.171
	10.007	0.936	0.868	1.000	0.237	0.027	0.456	0.120
	14.383	0.944	0.889	1.003	0.254	0.062	0.432	0.011
	18.025	1.000	0.938	1.061	0.039	−0.181	0.263	0.122
	21.146	1.023	0.966	1.083	−0.026	−0.262	0.197	0.098
	24.409	1.110	1.024	1.194	−0.382	−0.769	0.009	0.159

Table 6
Calibration parameter estimates and 95% confidence intervals on the square root scale for TEOM 30 °C samplers (type 16) as a function of temperature.

Site ID	°C	β			α			σ_i/β_i
		Estimate	Lower	Upper	Estimate	Lower	Upper	
1	−5.141	0.445	0.209	0.681	1.711	0.943	2.484	0.705
	0.731	0.772	0.653	0.893	0.712	0.308	1.110	0.261
	5.379	0.523	0.318	0.725	1.614	0.921	2.319	0.980
	10.143	0.711	0.521	0.900	1.031	0.383	1.678	0.608
	14.574	0.665	0.447	0.878	1.182	0.335	2.051	0.842
	17.908	0.568	0.324	0.806	1.639	0.698	2.602	1.057
	21.063	0.974	0.898	1.058	0.017	−0.351	0.355	0.103
	24.608	0.904	0.825	0.988	0.405	−0.074	0.862	0.236
2	−6.026	0.853	0.768	0.941	−0.114	−0.422	0.189	0.257
	0.547	0.942	0.857	1.028	−0.347	−0.653	−0.043	0.157
	5.344	0.967	0.890	1.053	−0.275	−0.566	−0.017	0.103
	10.016	1.069	0.959	1.177	−0.458	−0.808	−0.108	0.140
	14.554	1.138	1.016	1.261	−0.719	−1.143	−0.303	0.167
	17.999	1.100	0.985	1.205	−0.595	−0.989	−0.168	0.094
	21.114	0.724	0.552	0.882	0.639	−0.017	1.348	0.707
	24.333	0.924	0.838	1.011	−0.012	−0.465	0.435	0.334

Table 7
Calibration parameter estimates and 95% confidence intervals on the square root scale for the single FDMS TEOM sampler (type 19) as a function of temperature at Site ID 11.

°C	β			α			σ_i/β_i
	Estimate	Lower	Upper	Estimate	Lower	Upper	
−5.732	0.975	0.880	1.070	−0.189	−0.517	0.140	0.323
0.766	0.983	0.902	1.062	−0.076	−0.358	0.209	0.314
5.299	1.030	0.947	1.113	−0.101	−0.376	0.176	0.265
10.022	0.896	0.797	0.996	0.327	0.004	0.652	0.361
14.302	0.851	0.761	0.942	0.476	0.172	0.779	0.308
18.082	0.884	0.815	0.955	0.393	0.129	0.653	0.255
21.164	0.883	0.825	0.942	0.434	0.183	0.682	0.177
24.435	0.908	0.868	0.949	0.423	0.223	0.622	0.160

sampler. Notably, the estimated A and B for β are statistically significantly different from 1, with A being substantially less and B being slightly greater than 1. Analogously, the estimated A and B for α are statistically significantly different from zero with A being substantially greater and B being substantially less than zero. The random effect standard deviation $\sigma_t = 3.2799$ °C for α when compared with the fixed effect $T_{mid} = 15.7733$ °C indicates that there is substantial variation in the location of the sigmoidal curves across sites. Similarly, the random effect standard deviation $\sigma_s = 1.5321$ °C for β when compared with the fixed effect $S = 3.9384$ °C indicates that there is substantial variation in the scale or spread of the sigmoidal curves across sites. The standard deviations for the other random effects are relatively small. For α, the random error standard deviation for site 4 is estimated as 0.0385 (on the square root transformed scale). All of the variance structure estimates for sites 5, 7 and 10 are above 1 indicating that the random error standard deviations for these sites may be larger than for site 4. Only the standard deviation for site 10 is substantially greater than 1 and at minimum is likely to be at least twice the standard deviation for site 4. Analogously, for β, the random error standard deviation for site 10 is at least 25% higher than for site 4. The β for sites 5 and 7 may have lower random error than the β for site 4, but the confidence intervals for the variance factors include 1, so this is not conclusive. Finally, Table 11 shows the site-specific sigmoidal model parameter estimates for α and for β. Fig. 5 shows the typical sigmoidal model based on the fixed effects (grey curves), which is the same across all sites, and the site-specific models based on the random effects (added to the fixed effects) that vary across sites (black curves). In a mixed effects model, sites with more variable observations (in this case, parameter estimates), tend to *borrow* information from the other sites. For illustration, the calibration equations for site 4 are given by equations (5) and (6) where

$$\alpha(T) = 0.3755 + \frac{-0.3118 - 0.3755}{1 + e^{\frac{(19.5379+t)-T}{2.5413}}} \tag{10}$$

$$\beta(T) = 0.7240 + \frac{1.0474 - 0.7240}{1 + e^{\frac{(9.9925+t)-T}{6.0362}}} \tag{11}$$

Equations (10) and (11) should be compared to equations (7) and (8) that were derived for the same samplers using nonlinear regression and to the equations for the fixed effect model (which are not shown). We pooled together the TEOMs for sites 4, 5, 7, and 10 because they appeared to operate in a similar manner. By using the nonlinear mixed effects model, we *borrow* information from all of these sites in order to temper each site-specific prediction. The site-specific curve is *pulled* toward the *typical* curve, the curve determined by data from all sites. The *pull* will be stronger for sites with few observations and/or more variable observations. However, as shown in Fig. 5, the choice in which sites to group together as a sample from a specific population only makes much of a difference for the bias parameter α for site 10 given the greater variability in the observations for this parameter at this site.

6.3. TEOM imprecision

Although the primary focus of this study is calibration of the various samplers with the FRM sampler, it is useful to examine the imprecision of the samplers after adjusting the imprecision standard deviations (σ_i) for scale changes (σ_i/β_i). There were 9 TEOM samplers with six of these being 50 °C TEOMs. In general, the 50 °C TEOM imprecision standard deviation was higher than that of the FRM samplers. For most of the 50 °C TEOMs, the imprecision decreased with increasing temperature. The change in σ_i^2/β_i^2 with temperature was roughly linear as previously described in Bilonick et al. (2015). For the two 30 °C TEOMs and one FDMS TEOM, there is less clear evidence of a relationship with temperature.

7. Discussion and conclusions

The analysis presented here represents a complex, but necessary, step in incorporating data from a number of disparate measurement devices into a common epidemiology study. The $PM_{2.5}$ measurements made by the seven air monitoring programs in the 35-county area were independent efforts without an overall goal for integration or eventual use in a comprehensive health effects study. Making comparisons temporally and spatially (i.e., for purposes of characterizing variations in human exposure to ambient $PM_{2.5}$ throughout the region) requires measurements made effectively on the same scale. Fortunately, in the case of PARIES, enough multiple collocated measurements from different types of samplers were available so that all non-FRM samplers could be calibrated to the FRM sampler. We used the calibration equations reported above to convert all measurements made by non-FRM samplers to their FRM-equivalent values, and then used the resulting FRM-equivalent values in geostatistical modeling to develop daily ZIP-code-specific estimates of $PM_{2.5}$ exposure in the three-county study area. (Results of this exposure modeling will be presented in a future paper). This same approach is recommended for other retrospective epidemiology studies in order to take advantage of all available data without introducing exposure misclassification as a result of the relative biases between measurement devices. The general calibration methodology presented here could also be applied by air monitoring networks to develop robust calibration equations for the monitors in their operating region without requiring full collocated monitoring at every site. (As established above, at least three collocated monitors are required to conduct a properly designed calibration study at a single monitoring site).

It is noteworthy that the conclusions developed here for a number of $PM_{2.5}$ monitors and monitoring sites in the Pittsburgh region are generally consistent with those developed in Bilonick et al. (2015) for a single site (*42-3-8*, site 5 in Table 2). Specifically, (1) the 50 C TEOM samplers tended to show a large downward bias relative to the FRM sampler at low temperatures, and the magnitude of this bias decreased according to a nonlinear (sigmoidal) pattern with increasing temperature, (2) the other types of integrated samplers (i.e., speciation and dichotomous samplers) generally showed smaller biases relative to the FRM

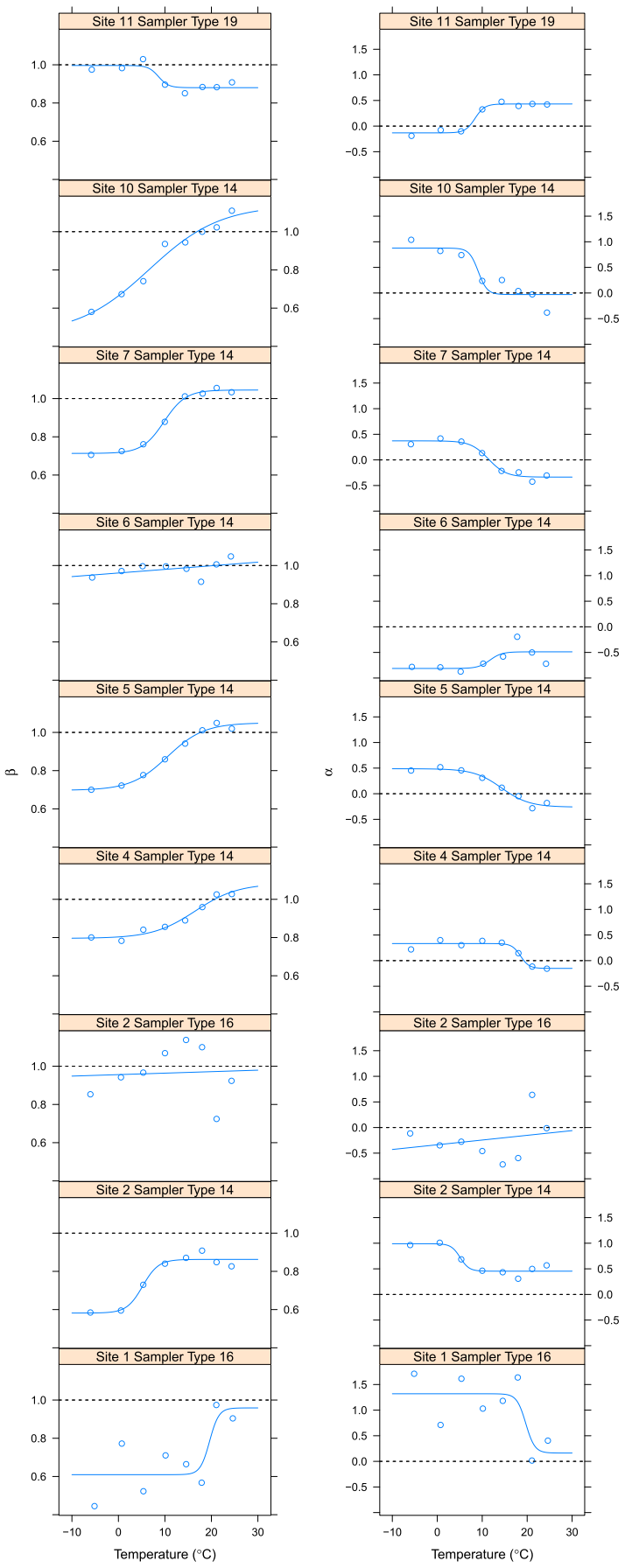


Fig. 4. Regression model fits for each TEOM. For β at Site 6 (Type 14) and Site 2 (Type 16), and for α at Site 2 (Type 16), linear regression was used. Otherwise, a nonlinear sigmoidal regression was used.

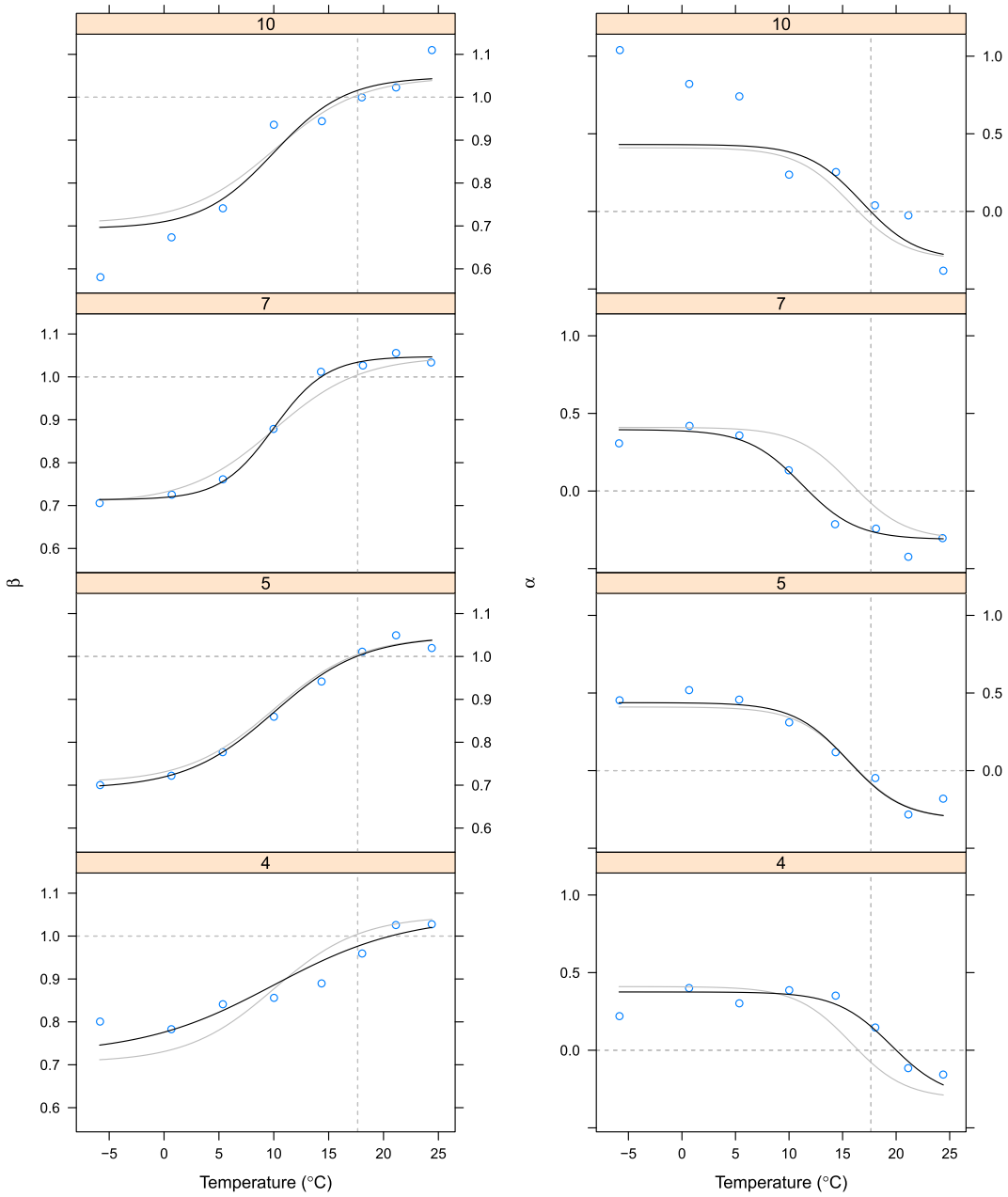


Fig. 5. Nonlinear mixed effects model sigmoidal fits to TEOM 50 °C (Type 14) calibrations parameters (β and α) as a function of temperature for sites 4, 5, 7, and 10. Fixed effect (typical) fit is denoted by the grey curve. Random effect (site-specific) fit is denoted by the black curve. The dashed horizontal line for β denotes the lack of a scale bias. The dashed horizontal line for α denotes the lack of a constant bias when there is no scale bias. The vertical dashed line for both β and α denotes the temperature (17.65 °C) at which the scale bias vanishes as estimated in Bilonick et al. (2015).

sampler that were not temperature-dependent, and (3) the FRM samplers tended to be more precise than the non-FRM samplers. (The noteworthy exception was the IMPROVE sampler, which had approximately the same imprecision as the FRM sampler). This consistency lends credence to our methodology.

Finally, it should be mentioned that the approach used here to model the temperature dependence of the TEOM was not ideal but was a necessary approximation to avoid substantially further complicating the analysis. Ideally, the effect of temperature would be directly incorporated into the structural equation model. However, incorporating nonlinear components into a structural equation model is complicated, whereas the stratification of measurements into eight temperature classes provided a workable alternative for our purposes. Also, because the temperatures that were used as the independent variable in the nonlinear regression and nonlinear mixed effects modeling were necessarily measured with some random error, the parameter estimates from these models (describing the temperature dependence of the relationship between TEOM and FRM) will have some bias. (As with ordinary regression, these models assume that the independent variable is a highly precise measurement having little or no random error). This bias should be relatively small, though, when the observations are close to the estimated site-specific curve.

Table 8
Sigmoidal model parameter estimates for $\beta(T)$ and $\alpha(T)$ for TEOM's using nonlinear regression models. Here A denotes the left asymptote and B denotes the right asymptote. T_{mid} denotes the location parameter and s denotes the spread or scale parameter on the temperature (°C) axis. Values shown in *italics* are fixed, not estimated. Rows with only estimates for A and B for either/or both α and β were estimated by ordinary regression with A denoting the intercept and B denoting the slope.

Site ID	Sampler ID	$\beta(T)$				$\alpha(T)$			
		A	B	T_{mid}	S	A	B	T_{mid}	S
1	16	0.609	0.959	19.607	*1.000*	1.320	0.162	19.687	*1.000*
2	16	0.956	0.001			−0.336	0.009		
	14	0.582	0.863	5.202	1.699	0.990	0.454	5.084	*1.000*
4	14	0.796	1.082	16.542	4.578	0.333	−0.152	18.524	0.951
5	14	0.697	1.050	10.572	3.934	0.488	−0.261	14.292	3.085
6	14	0.961	0.002			−0.813	−0.488	11.718	1.313
7	14	0.713	1.045	9.807	2.337	0.372	−0.336	11.378	1.965
10	14	0.464	1.139	6.586	7.635	0.876	−0.030	9.102	*1.000*
11	19	0.996	0.880	8.616	*1.000*	−0.132	0.432	8.395	1.080

Table 9
95% Confidence intervals for sigmoidal mixed effects model for TEOM calibration $\alpha(T)$. The variance factor is used as a multiplier × the random error standard deviation σ_ε for site 4.

	Fixed effects				Random effects				Variance factor			Site 4
	A	B	T_{mid}	S	σ_a	σ_b	σ_t	σ_s	Site 5	Site 7	Site 10	σ_ε
Lower	0.3411	−0.4029	11.9733	1.4008	0.0054		1.5197		0.3974	0.4312	2.2597	0.0202
Estimate	0.4099	−0.3118	15.7733	2.5413	0.0400		3.2799		1.0625	1.0792	5.3374	0.0385
Upper	0.4787	−0.2208	19.5733	3.6819	0.2940		7.0790		2.8408	2.7009	12.6067	0.0734

Table 10
95% Confidence intervals for sigmoidal mixed effects model for TEOM calibration $\beta(T)$. The variance factor is used as a multiplier × the random error standard deviation σ_ε for site 4.

	Fixed effects				Random effects				Variance factor			Site 4
	A	B	T_{mid}	S	σ_a	σ_b	σ_t	σ_s	Site 5	Site 7	Site 10	σ_ε
Lower	0.6799	1.0334	9.4662	2.0685	0.0028			0.4973	0.2501	0.1830	1.2669	0.0037
Estimate	0.7058	1.0474	9.9925	3.9384	0.0178			1.5321	0.5943	0.4504	3.0032	0.0071
Upper	0.7317	1.0614	10.5188	5.8083	0.1124			4.7204	1.4126	1.1088	7.1191	0.0139

Table 11
Predicted site-specific parameters for the nonlinear sigmoidal models fitted to the estimated calibration parameters for the eight temperature strata. The model predictions (black curves) are plotted in Fig. 5.

Site ID	Asymptote $t\to-\infty$ $A+a$	Asymptote $t\to\infty$ $B+b$	Location °C $T_{mid}+t$	Scale °C $S+s$
Sigmoidal Model for $\alpha(T)$				
4	0.3755	−0.3118	19.5379	2.5413
5	0.4378	−0.3118	15.5430	2.5413
7	0.3955	−0.3118	11.2089	2.5413
10	0.4309	−0.3118	16.8033	2.5413
Sigmoidal Model for $\beta(T)$				
4	0.7240	1.0474	9.9925	6.0362
5	0.6916	1.0474	9.9925	4.0302
7	0.7136	1.0474	9.9925	2.4034
10	0.6939	1.0474	9.9925	3.2836

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